

**a Training of capsid-specialized PLMs**

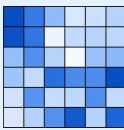
**Benchmark for pretrained PLMs**

| ESM2-650M                                                                                                                                                                                                                | ESM3-open                                                                                                                                                                                            | ESMC-600M                                                                                                                                                                                         | ProFluent-E1-600M |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <br>Three data splits <ul style="list-style-type: none"><li>Random split</li><li>Low-identity-held-out</li><li>Family-held-out</li></ul> | <br>Binary probes <ul style="list-style-type: none"><li>F1, precision, recall</li><li>False positive rate</li></ul> | <br>MLM&Perplexity <ul style="list-style-type: none"><li>Capsid proteins</li><li>Non-capsid proteins</li></ul> |                   |

ESMC-600M as base model

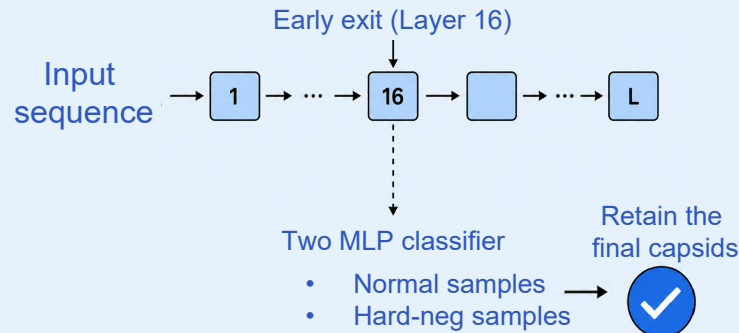
**ESMCapsid-S**  
Sensitive Capsid Detection

**LoRA+Supervised contrastive training**  
Capsid vs Non-Capsid protein

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- Mean-pooled embedding
  - MLP Classifier

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- Threshold calibration  
Optimize F1, constrain FPR

**Layer-wise probing and early-exit inference**

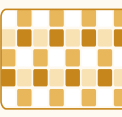


 **~2x faster**  
than full-depth inference

**F1 > 0.992**  
**FPR < 0.008%**

**ESMCapsid-C**  
Sensitive Capsid Detection

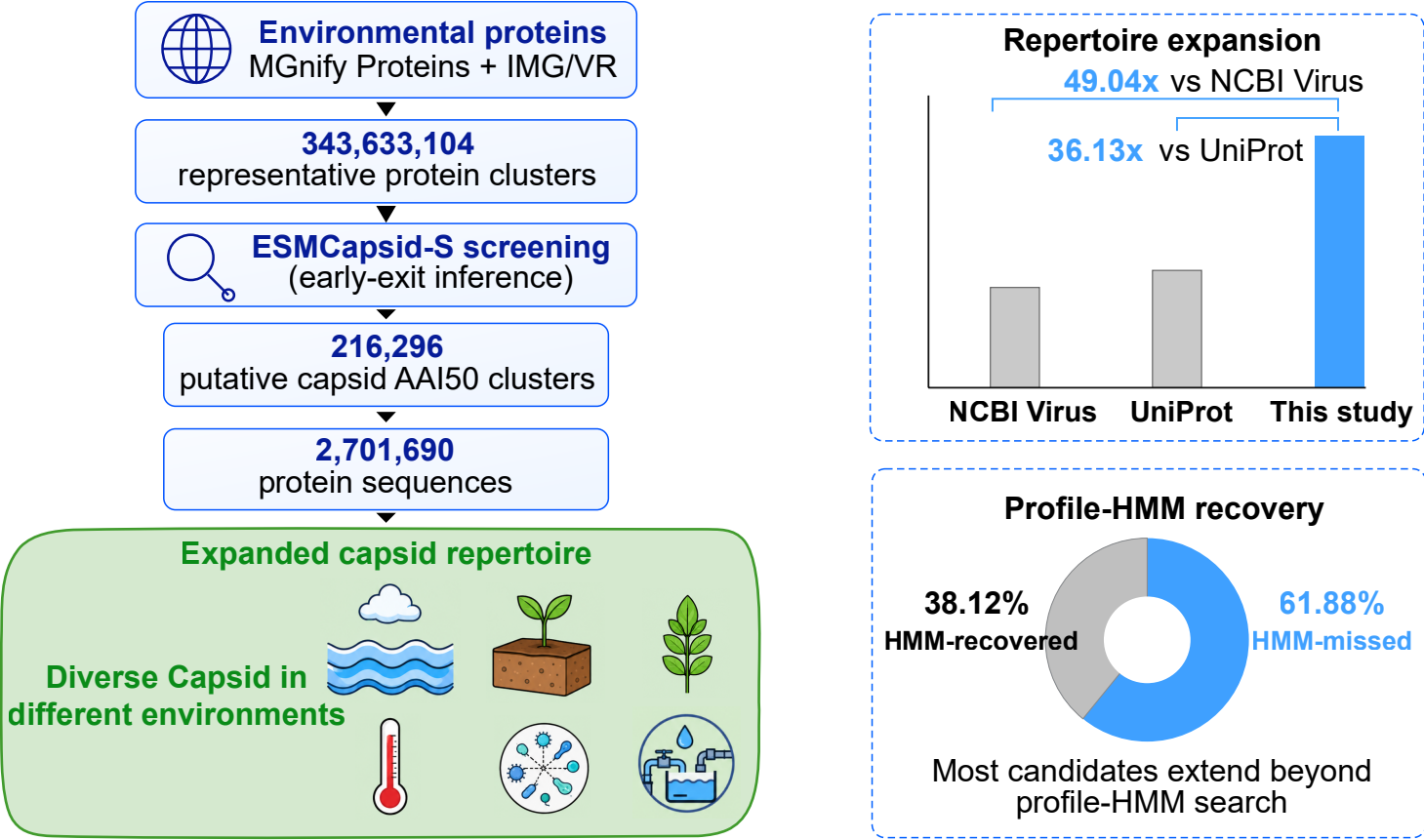
**Continual MLM pretraining**  
on redundancy-filtered capsid corpus  
>210,000 capsid proteins

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- MLM objective**
- 15% masking rate
  - 100,000 steps
  - AdamW

- 
- Improved embedding quality**
- MLM
  - Perplexity

- 
- Improved representation for**
- Taxonomy prediction
  - Capsid fold prediction
  - Genome type prediction
  - Host prediction
  - ...

**b Metagenomic discovery and expanded capsid repertoire**



**c SAE and semantic motif framework**

